

Training History Shapes Development

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Abstract

Prior training changes what a model is ready to learn next. In a controlled language-model environment, a source curriculum produces a large, selective advantage in acquiring a later capability, while matched controls remain near chance. The effect survives zero shared entity tokens and persists when the target is composed with a fixed derangement that blocks direct zero-shot answer transfer: source-trained models begin at chance yet subsequently learn substantially faster. We call this **developmental readiness**.

We then measure $V(S,D)$, the value of future corpus D from incoming model state S , and find State×Data interaction with corpus-ranking reversals across measured states. However, the stronger hypothesis that consequential developmental differences remain hidden from present behavior does not survive testing. In two prospectively specified experiments, behavior-matched states showed neither excess divergence under an identical future nor reliable differential preference between alternative futures. Internal gradient geometry clearly encoded training history, but did not reliably predict conditional data value, and a prospective temporal replay detected no localized transition in future learnability.

The resulting picture is narrower than our starting hypothesis: simple target competence can miss consequential readiness, while richer behavioral measurements capture more future-relevant variation than expected. Training history matters; identifying measurements sufficient to forecast its future consequences remains open.

1. Introduction

A central concern in the study of digital minds is that **behavioral evidence may underdetermine what kind of system we are observing**, and the concern extends across time: present behavior may not reveal how a model will change under future experience. Rather than assuming that behavioral similarity either does or does not imply similarity of underlying state, we test this experimentally by asking whether behaviorally matched checkpoints respond differently to controlled future experience. What we find is more nuanced than a yes or a no. Training history changes future learnability a great deal, strongly enough that a single competence score misses it entirely, while a richer behavioral measurement captured more of that variation than we expected. For AI safety, this matters because evaluations at one point in training may not fully characterize how a model will respond to later fine-tuning, continued pretraining, or deployment experience. Throughout, we use **developmental state** to mean *history-dependent properties of a checkpoint that affect its response to later training*; we do not assume it corresponds to consciousness, identity, or welfare.

We began by asking whether pairwise transfer relations could be composed into a useful training order. Repeated failures (non-composing transfer effects, interference under sequential curricula, and unreliable prospective prediction) suggested that data value could not be treated

as a fixed property of a corpus pair. This motivated $V(S,D)$, the value of future data D conditional on what the model has already become.

Our contributions are: **history-dependent developmental readiness**, shown on a target whose answer is unavailable to the source model beforehand, so what transfers is not the answer; **controls against simpler explanations**, since the effect is selective, survives zero shared entity tokens, and persists across $32\times$ capacity; **state-dependent future data value**, with $\text{State}\times\text{Data}$ ordering reversals that state-aware selection nonetheless fails to exploit; and preregistered tests showing that **the stronger hidden-state story did not survive**.

The result is a sharper question than the one we started with. Developmental readiness is real, consequential, and invisible to a single capability measure, yet the measurements that captured it were behavioral, not internal. The open problem is therefore: **which present measurements are sufficient to forecast what a model is ready to become?**

2. Related Work

Classical curriculum learning established that ordering examples can alter optimization and generalization in non-convex models (Bengio et al., 2009). For language models, Skill-It formalizes ordered skill relations and uses prerequisite structure for more data-efficient training (Chen et al., 2023), and Procedural Pretraining shows that front-loading abstract structured data materially accelerates later acquisition of natural language, code, and mathematics (Jiang et al., 2026).

A separate literature finds that learning is often stagewise, with internal development preceding visible behavioral change: loss-landscape degeneracy drives distinct developmental stages in transformers (Hoogland et al., 2025), and training history can remain directly readable, as activations linearly encode training-order recency (Krasheninnikov et al., 2025). Mechanistic work connects induction-head emergence to a sharp transition in in-context learning (Olsson et al., 2022). Together these motivate a prerequisite-style framing, in which data value is approximately global or pairwise; we ask instead whether the same future corpus has different marginal value from different incoming model states.

A parallel literature already treats data value as model-dependent. Datamodels predict counterfactual outputs under training-set changes (Ilyas et al., 2022), DoReMi optimizes a global domain mixture (Xie et al., 2023), MATES adapts a learned data-influence model to the *evolving* pretraining model and selects data predicted most useful at the current stage (Yu et al., 2024), and Gu et al. (2025) formulate data selection as optimal control over training dynamics. We therefore do **not** claim that state-dependent or model-aware data value is itself novel; that territory is well covered. Our narrower contribution is methodological: controlled continuation measurement of $V(S,D)$ under *manipulated* developmental histories, a direct test of whether present behavior or internal telemetry predicts those future-learning consequences, and the requirement that state-aware selection beat a strong state-blind baseline before any adaptive claim is made.

On the digital-minds side, Perez & Long (2023) argue self-reports could eventually bear on morally significant internal states while stressing that present reports are often spurious, and persona-vector work connects training history to future behavioral change through readable

internal variables (Chen et al., 2025). Controlled continuation experiments offer a complementary external assay: rather than relying on self-description, they test whether behavioral similarity hides consequential differences in how a model responds to future experience.

3. Methods

3.1 Controlled source histories and targets

We train small decoder-only language models from scratch on synthetic language-like corpora under next-token prediction, varying source history while holding architecture, token budgets, and downstream data fixed. Every stream shares one surface template, the `<entity>` is `<value>` ., so no stream is identifiable from surface form; only the *relationship* differs, giving a ground-truth record of developmental history.

Source conditions are **A**, training that encourages stable entity–value retrieval/binding; **A'**, a control preserving surface statistics while disrupting that structure; and **BG**, background training. All arms then receive the same target continuation.

Targets: **B** is a binding/retrieval task with resampled identities, so answers cannot be memorized as fixed associations. **B₂** is the centerpiece: retrieval composed with a **fixed derangement**, so the retrieved value is never the correct answer. This **blocks direct transfer of the target answer**, and we confirm it blocked, with every arm at or below chance before continuation. It does not block transfer of procedure or structure, which may be part of what readiness consists of. **C** is a learnable parametric-association negative control that does not require contextual retrieval.

3.2 Readiness metrics, controls, and prospective specification

We report $t=0$ performance before any target training, AULC, final performance, and rate-only (AULC - t_0). The decomposition matters because head-start and rate effects move differently: an advantage in rate-only while $t=0$ sits at chance **supports a readiness interpretation rather than direct answer transfer**. Controls: a content-disjointness condition uses zero shared entity identities; specificity is tested against C, interpreted only once BG shows C is learnable; and we replay the core contrast at roughly 1×, 8×, and 32× non-embedding capacity.

Discovery, calibration, and confirmation seeds are separated; for the prospective experiments, protocols, analysis plans, and selection artifacts were **preregistered and hashed before the corresponding outcomes were evaluated**, and compound gates fail if any criterion fails. This changed conclusions: a causal ablation returned a null interaction that we report as **inconclusive** rather than negative, because an efficacy check showed the intervention never measurably reduced the capability it targeted.

3.3 State-conditioned data value, prospective tests, and internal readout

$V(S,D)$ is the measured learning value of continuing from saved state S on corpus D , under an identical continuation protocol with fresh optimizer state, isolating information carried in the weights. We build a balanced matrix of incoming states $\times$ candidate corpora and decompose

variance into state main effect, data main effect, and State×Data interaction, whose strongest qualitative signature is an **ordering reversal**, $\operatorname{argmax}_D V(S_1, D) \neq \operatorname{argmax}_D V(S_2, D)$. A selector must beat both random and the **state-blind global-best corpus**; beating random alone is achievable by learning the data main effect. Validation is leave-one-**state**-out, never row-wise, which would leak a state's value profile through its other corpora.

Three **prospective continuation tests** follow, each giving matched or dense states controlled future experience. **P1** asks whether behaviorally matched states diverge under a *common* future, selecting pairs on present observables only and freezing the pair list before any continuation. **Fork** asks whether they prefer *different* futures, branching both members onto two corpora chosen for the strongest prior ordering reversal under a preregistered difference-in-differences estimand. **P2** asks whether future learnability changes at a localized point in training, measuring $V(S_t, B)$ directly rather than competence across a preregistered source window.

Separately, **E4** is an **internal-state readout** experiment rather than a continuation test: for each saved state and candidate corpus, on identical future minibatches, we measure the geometry of the gradient that candidate data induces on current weights (norms, layerwise gradient-mass entropy, and gradient-weight and cross-corpus alignment), and ask whether those measurements predict $V(S, D)$.

4. Results

The experiments progressively narrowed the claim: training history robustly changes readiness, but stronger claims about hidden behavioral state, temporal staging, predictive internal readout, and adaptive control did not survive testing.

Question	Result
Does history change future learnability?	Supported
Is it generic acceleration?	Not supported , control capability unmoved
Is it memorized content reuse?	Ruled out , survives zero shared tokens
Is it target competence, or readiness?	Supports readiness , zero-shot target answer blocked, learning still accelerated
Does it survive scale?	Exploratory support , 32×
Does future data value depend on state?	Supported , interaction with ordering reversals
Do behavior-matched states diverge under one future?	No excess divergence detected , 71-pair null (P1)
Do behavior-matched states prefer different futures?	No reliable difference detected (Fork)
Is there a localized transition in future learnability?	No localized change detected in the preregistered window (P2)
Does internal geometry identify history?	Supported
Does it predict $V(S, D)$?	Inconclusive
Can we choose the best next corpus yet?	Not supported , state-aware selector loses to global-best

4.1 Developmental readiness

The centerpiece is B_2 , the deranged target. Because the derangement blocks the retrieved answer, every arm begins at or below chance: A scores 0.0122 (disjoint surface) and 0.0156 (shared) against a chance floor of 0.0156, so the target answer is unavailable before continuation. The source arm nonetheless learns far faster: a rate-only advantage over the matched control of **+0.5438** on the disjoint surface and **+0.3967** shared, reaching final accuracy **1.0000** against A' at 0.26/0.51 and BG at 0.19/0.32, across 24 units (4 seeds $\times$ 3 arms $\times$ 2 surface conditions). Per the preregistered reading table, a rate-only advantage with $t=0$ at chance **supports a readiness interpretation rather than direct answer transfer**. The evidence is real but not overwhelming ($n = 4$ per arm, effect/noise 1.6–1.9, control-arm sd 0.14–0.21).

On the ordinary target B, as supporting evidence, A reaches 0.1322 ± 0.0149 at $t=0$ against 0.0156 ± 0.0087 (A') and 0.0156 ± 0.0051 (BG), roughly 8.5 $\times$ chance, with controls at the floor, and $AULC(A) - AULC(A') = +0.5265$.

Prior training buys readiness to learn, not the answer

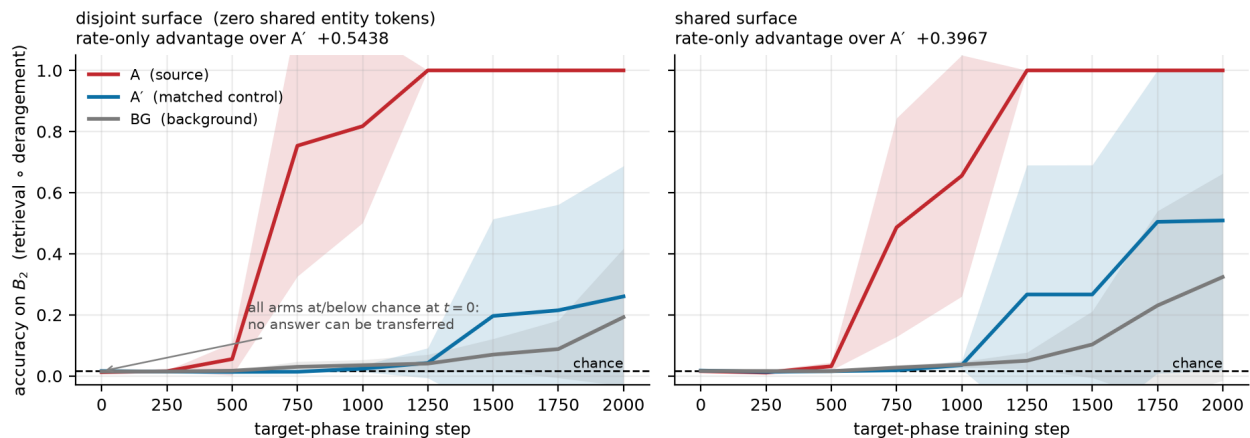


Figure 1. Acquisition of B_2 (retrieval $\cdot$ derangement), mean $\pm$ sd over four seeds per arm. The derangement keeps zero-shot target performance at chance, blocking direct transfer of the target answer. The source-trained arm nevertheless acquires the capability substantially faster, and the rate-only advantage is at least as large when source and target share zero entity tokens (left).

4.2 Specificity, disjointness, and scale

The effect is not generic acceleration: on the negative control C the advantage is +0.0395, roughly thirteen times smaller than the B effect. This null is meaningful because C is learnable: BG reaches 0.410 against a prespecified 0.30 competence gate. It is not memorized content: with zero shared entity tokens the head start is fully retained (0.1567 disjoint vs 0.1431 shared, 111% retention). It persists with capacity: zero-shot B advantage of 0.126 ± 0.016 at 1 $\times$, 0.157 ± 0.042 at 8 $\times$, and 0.097 ± 0.037 at 32 $\times$, with controls remaining near chance on zero-shot B throughout, though these runs are exploratory rather than a powered scaling study.

4.3 State $\times$ Data interaction exists, but is not yet actionable

In the audited balanced matrix (13 complete states $\times$ 3 candidate corpora), continuation value depends on the pairing of incoming state and future data. Corpus rankings reverse across states, though the effect is highly imbalanced: FACT is optimal for 12/13 states and BIND for BG-500.

Corpus value is therefore not strictly state-independent over the measured states, but a strong global corpus advantage remains.

A leave-one-state-out selector using state information beats random selection (mean regret 0.0473 vs 0.0758) but performs substantially worse than the state-blind global-best baseline (0.0207; top-1 10/12 vs 11/12). The current predictor therefore captures broad corpus quality better than the conditional structure needed for adaptive selection. We establish State×Data variation over these unmatched states, but do not establish that our present state representation can exploit it.

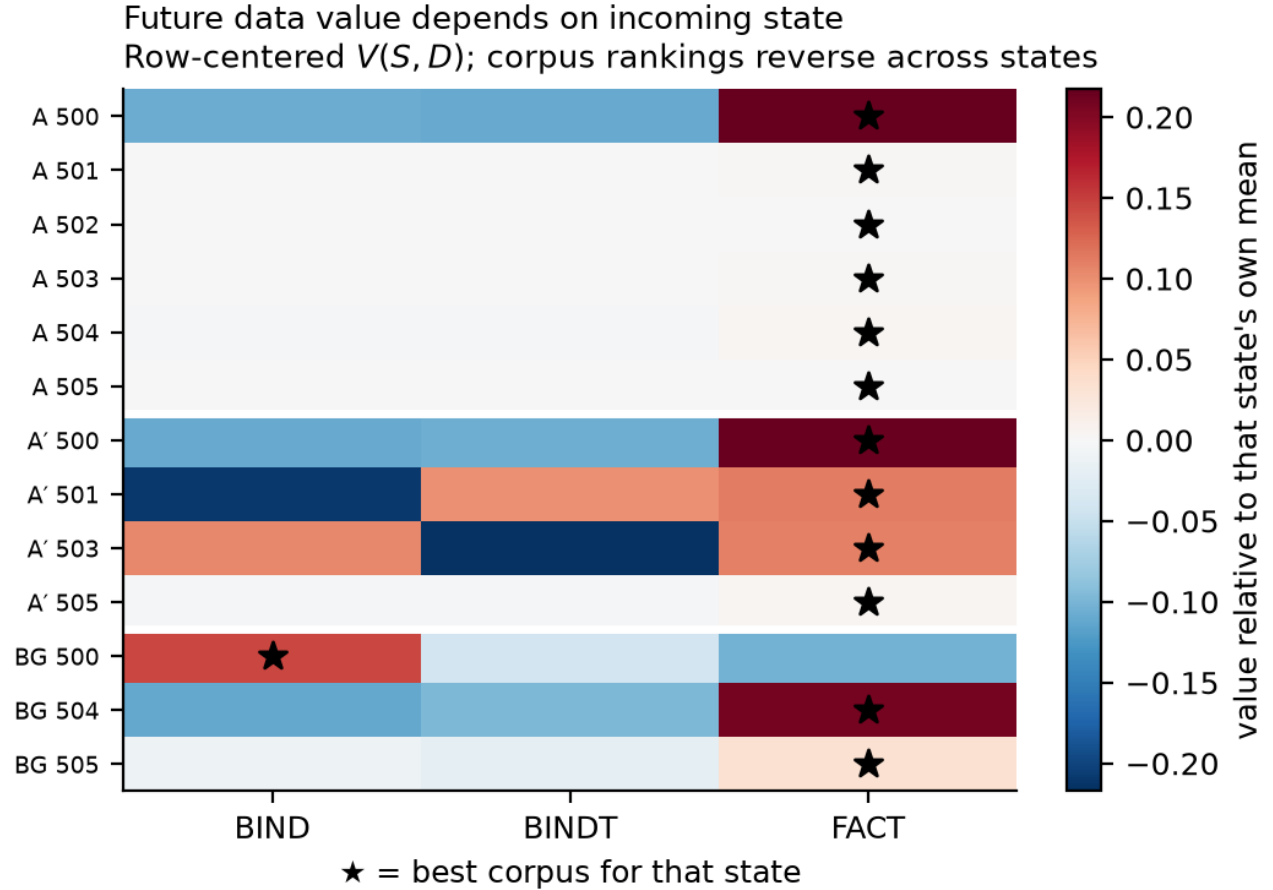


Figure 2. Row-centered $V(S, D)$, so colors compare candidate corpora within each incoming state rather than absolute value across states. Stars mark the best corpus for each state. FACT is optimal for 12/13 states, while BIND is optimal for BG-500, demonstrating a ranking reversal but also a strong global corpus advantage. Consistent with this imbalance, state-aware selection does not outperform the state-blind global-best baseline.

4.4 Boundaries and negative results

No excess divergence under a common future (P1). Pairs matched on present observables, frozen before any outcome existed, do not diverge more than unmatched same-arm pairs when given identical continuations: final 0.2893 vs 0.2700 (permutation $p = 0.704$) and rate-only 0.1325 vs 0.1496 ($p = 0.504$), over 71 matched pairs against a 376-pair null. A significant $t=0$ result is circular rather than a finding, since zero-shot accuracy is itself part of the matching vector; the metrics *not* matched on show nothing. This is the better-powered of the two matched-state tests.

No reliable differential future preference (Fork). Matched states might respond alike to one future while differing in *which* future suits them. Forking 16 frozen pairs onto the two corpora with the strongest prior ordering reversal gives an interaction of +0.0888, 95% CI [-0.1173, +0.3162], spanning zero, with 8 of 16 pairs sharing the aggregate sign, exactly chance. Reversals do occur; what we cannot detect is a *reliable* differential preference. At 16 pairs the interval is wide enough that this test mainly fails to add evidence rather than subtracting it.

No localized change detected in the tested window (P2). An exploratory changepoint fit on zero-shot competence had suggested a localized early change. Because competence is not learnability, we tested prospectively: 48 checkpoints across source steps 150–450, each given an identical continuation. Held-out comparison gives linear 0.1333, sigmoid 0.1416, changepoint 0.1425; linear wins by 5.8%, inside the preregistered not-distinguishable band, and $V(S_p, B)$ correlates with source step at only $r = -0.199$. The original changepoint was fitted to traces sampled every 250 steps and carries no better than 250-step resolution; per the frozen protocol the window was not re-centred. This bounds detection within the tested window rather than establishing that no transition exists.

Mechanism, readout, and control. Our preregistered induction-style mediator failed its confirmatory gate on fresh seeds; a later retrieval statistic separates A from A' without cleanly replicating against BG, so we treat it as history-associated rather than A-specific, and the necessity test is inconclusive because the intervention never reduced the capability. Gradient geometry separates histories cleanly: gradient norm on B of 0.54 against 0.29 for the matched control, $\cos(\nabla B, \nabla C)$ of +0.73 against +0.19, yet as a *predictor* of $V(S, D)$ it is **inconclusive and not promoted**: the two prespecified objectives disagree in direction, both bootstrap intervals touch zero, and only 13 states have both geometry and a complete $V(S, D)$ row. This is the core boundary of the paper: **identifying which history produced a state is not the same as predicting the future consequences of that state**, and it is why the prospective what-next tournament remains gated and unrun. Full gate criteria, outlier analyses, and per-experiment archaeology are in RESULTS.md.

What bounds the claim: four negative and boundary results

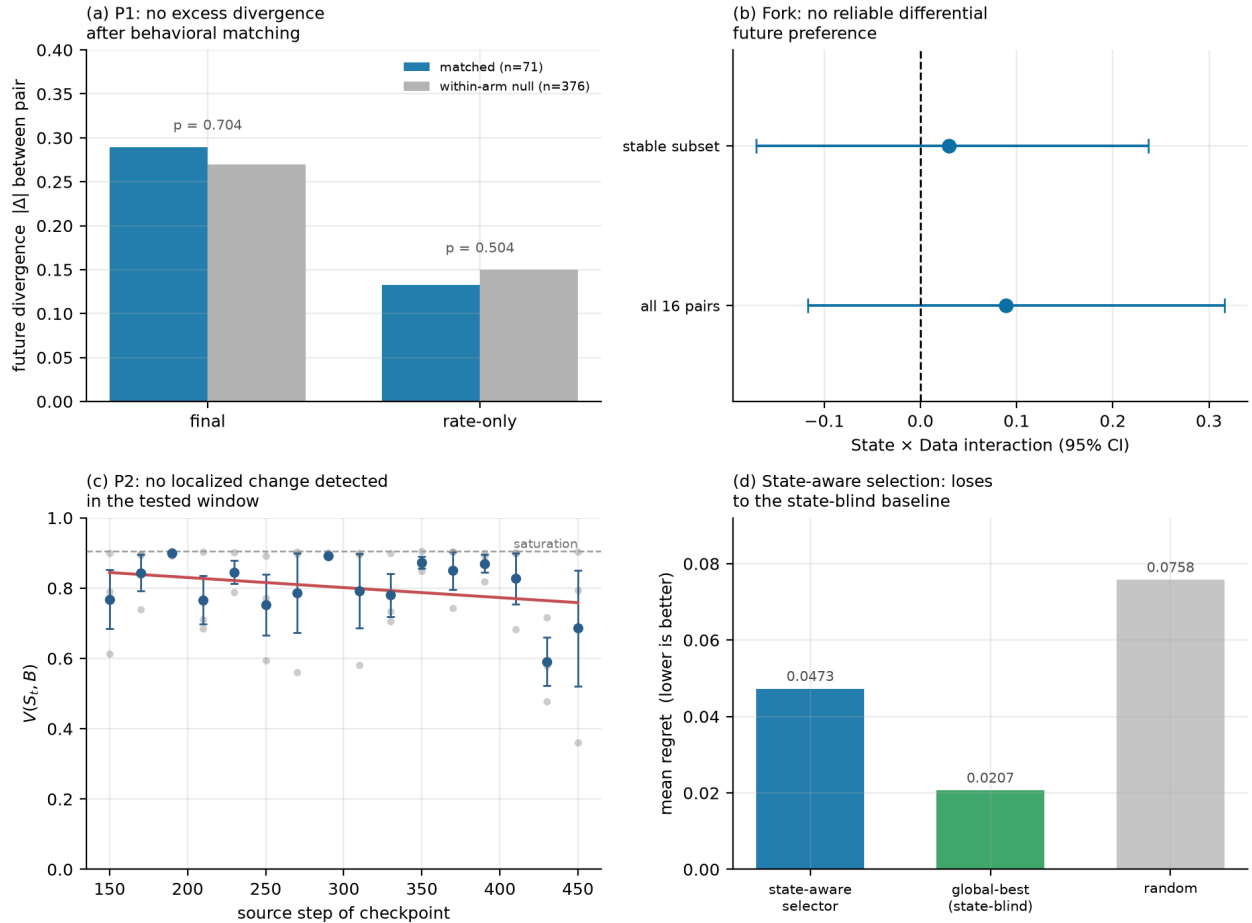


Figure 3. The four results that bound the claim. **(a)** P1: no excess divergence after behavioral matching, on either informative metric. **(b)** Fork: the State×Data interaction interval spans zero. **(c)** P2: no localized change detected in the tested window; note that all 48 continuations reach ceiling on the target, so the assay has limited dynamic range over much of this window. **(d)** state-aware selection loses to the state-blind global-best baseline.

5. Discussion and Limitations

We began by asking whether training order could be reduced to a static curriculum relation. Our experiments pushed us beyond one: past data changes the learner, and therefore changes the value of future data, a shift the results forced rather than one we assumed. The State×Data interaction suggests that the effect of future training data cannot always be treated as intrinsic to the corpus alone; its consequences depend on the model state receiving it.

The central synthesis is about which measurements capture what. A single capability score badly underestimates readiness: on B_2 the source model sits at chance zero-shot yet acquires the capability far faster than either control, even across disjoint content. Judging it by what it can currently do would miss the difference entirely. Yet the *richer* behavioral vector we matched on, zero-shot accuracy and loss across two capabilities, was enough that neither prospective test detected reliable residual divergence. These carry different weight: P1, at 71 pairs against a 376-

pair null, is well powered; the Fork, at 16 pairs, is small enough that its wide interval mainly fails to add evidence rather than subtracting it. The formulation we can defend is narrow: **developmental readiness can be invisible to a single capability measure without being deeply hidden from behavior in general.**

This reframes the digital-minds question from *does behavior hide developmental state?* to **which present measurements are sufficient to forecast future plasticity?** We do not show that behaviorally similar systems harbour hidden future divergence; we show that here a modest multi-capability profile already accounted for it where a single competence score did not. The experiments illustrate both directions of epistemic error relevant to AI welfare: current competence can understate consequential differences in future plasticity, while internal differences that reliably encode history can still overstate what we know about consequential latent state. Read together, the four instruments (target competence, richer behavioral profiles, internal geometry, and future-learning assays) capture different properties of the same checkpoints: history proved internally readable without being predictively actionable, while simple competence missed readiness that richer behavior largely captured.

Two consequences follow. Capability evaluation and update-risk evaluation are distinct problems: a model can lack a capability now while differing substantially in how readily it will acquire that capability later. And if training history changes what a model is ready to learn, identical post-training or deployment experience may have different behavioral consequences depending on the incoming checkpoint; in safety-relevant settings that is a potential mechanism for alignment drift, although alignment drift itself is not tested here. The same distinction bites internally: gradient geometry cleanly identifies which history produced a state, and that is **history recognition, not prediction of future consequences.** The trajectory this work traces (static curriculum → state-conditioned value → predictive state readout → adaptive control) is therefore complete only through its first two stages.

Limitations. "Developmental state" is operational, not uniquely identified: nothing here establishes a low-dimensional latent coordinate, a discrete stage, or Markovian dynamics. The behavioral-sufficiency result is a boundary condition, not a universal claim: it holds for *this* matching vector, microworld, scale, and set of futures, and a richer future set, coarser vector, different substrate, or larger model could reverse it. The synthetic environment is what makes the controls possible and is also the clearest gap between this setting and the phenomena the digital-minds framing cares about. Breadth remains unresolved: disjoint identities rule out content memorization, but the effect may still be specific to a related computational family. Technical caveats compound this: scale evidence is exploratory, mechanism is unresolved, prediction was not achieved, $V(S,D)$ continuations reset optimizer state and so measure information in weights rather than full training state, and the P2 assay's limited dynamic range bounds what it could have detected rather than showing that no temporal structure exists.

Dual-Use and Ethical Considerations

Most experiments were conducted on small, simple models and synthetic tasks designed to expose training dynamics directly. Sensitive or welfare-relevant topics were not required to elicit history-dependent differences in later learning, and the study does not rely on conversational self-report or preference elicitation. Instead, we construct the relevant training histories directly and intervene on prior experience while holding architecture and subsequent

training fixed, providing a ground-truth record of developmental history and a causal test of whether that history changes later learnability. We therefore establish a causal relationship between training history and future learning dynamics in this controlled setting. We do **not** establish ground-truth preferences, introspective access, subjective experience, moral status, or a uniquely identified internal developmental state.

Future Work

The next question is whether developmental readiness remains behaviorally legible in semantically richer environments. Our microworld removes connotation to isolate learning dynamics, but early experience may establish latent dispositions that become meaningful only when later experience supplies the relevant context. This motivates a **history** $\times$ **environment** test: vary developmental history, hold the later semantic environment fixed, and ask whether identical experience elicits different preferences, interpretations, or behavioral tendencies. Such a setting also provides a harder test of our behavioral-matching nulls, since a modest capability profile may be sufficient in a connotation-free microworld but not where later experience can activate dispositions laid down earlier. In parallel, adaptive control requires a state representation that predicts held-out $V(S,D)$, adds information beyond behavior, and beats a global-best baseline. That experiment will also need a **collapse-robust** outcome statistic: every P2 continuation reached ceiling on the target and some then destabilised, so a bare final-probe score partly measures where an instability happened to fall rather than the value of the data, and could let a readout post respectable regret by predicting optimiser noise. Only after those conditions are met should prospective what-next selection be attempted. Toward that end we have implemented a small prospective feasibility prototype of the loop: fresh states, readouts fused at the training endpoint, a complete $V(S,D)$ matrix, and predictions frozen before held-out outcomes are revealed. It exists to check that the experimental loop closes without leakage, not to supply evidence, and no results from it are reported here. The same assay would also support an introspection benchmark, extending controlled ground truth into model-identity research: ask whether a model can predict its own future learning value $V(S,D)$, compare that self-prediction against external behavioral and mechanistic predictors, and score all three against the actual continuation outcome. Nothing of the kind is demonstrated here.

6. Conclusion

Training history changes not only what a model can do, but what it is ready to learn next. We find a large, selective readiness advantage that survives disjoint content and appears even when the target capability is absent before training, while future data value varies with incoming model state. Yet the stronger hidden-state story did not survive: behavior-matched states showed no reliable residual divergence, no localized transition in future learnability was detected, and internal geometry that encoded history did not reliably forecast conditional data value. The resulting distinction is sharper than our starting hypothesis: present competence can miss consequential readiness, but those consequences may be more behaviorally legible than expected. The open problem is therefore not merely to detect training history, but to identify measurements sufficient to predict how a model will respond to future experience.

Code, Data, and Licensing

- **Code repository:** <https://github.com/walmsley-lab/ml-innovation-3-interpretability-sprint>
- **Data:** synthetic corpora are generated deterministically from the code and recorded seeds/configurations.
- **Reproducibility:** experimental protocols, analysis plans, configurations, and selected frozen artifacts are version-controlled in the repository. The code and recorded seeds support reproduction of the main experiments and analyses. For prospective experiments, key analysis and selection decisions were frozen before evaluating the corresponding outcomes.
- **Licensing:** source code is released under the **Apache License 2.0**; this report and other written/figure content are released under **Creative Commons Attribution 4.0 International (CC BY 4.0)**.

Author Contributions

Patrick Walmsley designed and carried out the experiments, audited the resulting claims, and wrote the report independently.

This work was carried out during a sprint alongside others in the SF Bay Area. Kryisia Koneni, Trevor Harrison, Augustus, and Patrick Walmsley acted as sounding boards for one another throughout.

AI was used heavily in experimental design and execution; see Appendix B.

References

1. Bengio, Y., Louradour, J., Collobert, R., & Weston, J. (2009). **Curriculum Learning**. *Proceedings of the 26th International Conference on Machine Learning*, 41–48.
2. Chen, M. F., Roberts, N., Bhatia, K., Wang, J., Zhang, C., Sala, F., & Ré, C. (2023). **Skill-It! A Data-Driven Skills Framework for Understanding and Training Language Models**. arXiv:2307.14430.
3. Chen, R., Ardit, A., Sleight, H., Evans, O., & Lindsey, J. (2025). **Persona Vectors: Monitoring and Controlling Character Traits in Language Models**. arXiv:2507.21509.
4. Gu, Y., Dong, L., Wang, H., Hao, Y., Dong, Q., Wei, F., & Huang, M. (2025). **Data Selection via Optimal Control for Language Models**. *ICLR*. arXiv:2410.07064.
5. Hoogland, J., Wang, G., Farrugia-Roberts, M., Carroll, L., Wei, S., & Murfet, D. (2025). **Loss Landscape Degeneracy Drives Stagewise Development in Transformers**. *TMLR*. arXiv:2402.02364.
6. Ilyas, A., Park, S. M., Engstrom, L., Leclerc, G., & Madry, A. (2022). **Datamodels: Predicting Predictions from Training Data**. *ICML*. arXiv:2202.00622.
7. Jiang, L., Shinnick, Z., van den Hengel, A., Saratchandran, H., & Teney, D. (2026). **Procedural Pretraining: Warming Up Language Models with Abstract Data**. arXiv:2601.21725.
8. Krasheninnikov, D., Turner, R. E., & Krueger, D. (2025). **Language Models' Activations Linearly Encode Training-Order Recency**. arXiv:2509.14223.
9. Olsson, C., Elhage, N., Nanda, N., et al. (2022). **In-context Learning and Induction Heads**. arXiv:2209.11895.
10. Perez, E., & Long, R. (2023). **Towards Evaluating AI Systems for Moral Status Using Self-Reports**.

arXiv:2311.08576. 11. Xie, S. M., Pham, H., Dong, X., Du, N., Liu, H., Lu, Y., Liang, P. S., Le, Q. V., Ma, T., & Yu, A. W. (2023). **DoReMi: Optimizing Data Mixtures Speeds Up Language Model Pretraining**. *NeurIPS* 36, 69798–69818. arXiv:2305.10429. 12. Yu, Z., Das, S., & Xiong, C. (2024). **MATES: Model-Aware Data Selection for Efficient Pretraining with Data Influence Models**. *NeurIPS* 37, 108735–108759. arXiv:2406.06046.

Appendix

A. Claim-status summary

Statuses use four distinct labels, because a null, a rejected hypothesis, and an ineffective intervention do not license the same conclusion. **Supported**; **Ruled out** / **Rejected by valid test**, a test with demonstrated power returned against the claim; **No reliable evidence detected** / **Not supported**, a well-executed test found nothing, which bounds rather than refutes; **Inconclusive** / **Not established**, the test could not adjudicate. The progression is: phenomenon established → simple explanations ruled out → stronger hidden-state hypotheses constrained → predictive readout unresolved.

Established

Claim	Status
Training history produces a selective future-B advantage	Supported
The effect is readiness rather than task transfer (B ₂)	Supported
State×Data interaction exists	Supported
Gradient geometry identifies training history	Supported
Effect persists across 32× capacity	Supported (exploratory)

Boundaries and rejected explanations

Claim	Status
The effect is generic learning acceleration	Not supported ; specificity control near-null
The effect is simple entity memorization	Ruled out by disjoint-content control
An off-distribution induction mediator M explains the effect	Rejected by confirmatory test
Behavior-matched states diverge under an identical future (P1)	No excess future divergence detected (71 matched pairs)
Behavior-matched states prefer different futures (Fork)	No reliable differential future preference detected (16 pairs)
Future learnability changes in a localized window (P2)	No localized change detected in the pre-registered window

Unresolved

Claim	Status
The retrieval statistic is an A-specific causal mediator	Not established
The top-4 ablation adequately tests necessity	Inconclusive ; intervention ineffective
Gradient geometry predicts conditional data value $V(S,D)$	Inconclusive ; objectives disagree at $n = 13$
Current telemetry can steer training	Not supported ; fails the global-best baseline
The prospective adaptive tournament is licensed	Not licensed by current evidence

B. What the model sees

All streams share the template the `<entity>` is `<value>` .; only the relation differs. In **BIND** the queried entity appears earlier and the answer must be retrieved from context. In **FACT** it does not appear earlier and the answer is a globally fixed association held in the weights. In **BINDT** (B_2) the answer is a fixed derangement of the bound value, so retrieval alone gives the wrong token.

Across 256 zero-shot BIND prompts, with no target-phase training:

history	exact answer correct	prediction is a value from the context
A	0.113	1.000
A'	0.008	0.133
BG	0.004	0.074
<i>chance</i>	0.016	0.109

The second column is the more mechanistic one: restricting the answer to values appearing in the context *is* the retrieval behavior, and it separates histories far more sharply than exact accuracy.

C. LLM Usage Statement

LLMs were used extensively, in distinct roles. **Claude** for implementation: wrote and debugged the experimental code, orchestrated runs across cloud workers, and built the analysis and audit tooling. **ChatGPT** for reasoning through experimental design: pressure-testing hypotheses, sequencing experiments, and interrogating what each result could and could not support. **DeepSeek**, used sparingly for inspiration and review. **Kimi K3** and **Perplexity** for literature search.

Experimental protocols, frozen gates, raw outputs, and final claims were checked against generated artifacts and independent audit scripts. LLM suggestions were not treated as evidence; claims were promoted only when supported by the corresponding experiments, and several LLM-proposed framings were discarded when the frozen criteria did not support them.